

Name:	B/o Praveena	Age: 14 days	Sex: Male
Sampling date: 28/09/2022		Received on: 30/09/2022	
Ref Dr:	Dr. D V Suresh, MD	File # : 1146/22	
Hosp/ City:	SKS Hospital, Salem		

Biotinidase Enzyme Activity

Date: 01/10/2022

Method: Wolf's colorimetric method.

Sample: Serum

Result	Serum: nmol / min / ml Normal : 4.5 – 7.5 – 12.0 Partial : 2.5 – 4.5 Deficient : Less than 2.5	Dried blood spots: OD Deficient: OD < 0.010 Partial : OD:0.010- 0.030 Normal: OD > 0.030	Impression
1 st Reading	5.88 nmol / min / ml	-	
2 nd Reading	5.20 nmol / min / ml	-	
Average	5.54 nmol / min / ml	-	NORMAL

Impression: Normal Biotinidase enzyme activity.

- Elevated Biotinidase enzyme activity may be seen in Glycogen Storage Disorders.
- Normal Biotinidase enzyme activity does not rule out Biotin deficiency due to other reasons like dietary deficiency, absorption defects [malabsorption syndrome], SMVT gene defects or high intake of raw eggs etc.
- Classical Severe Biotinidase enzyme deficiency should have activity less than 5 % of normal. Activities up to 10 % are also considered deficient.
- Partial deficiency is generally not considered to be symptomatic unless associated with some other stressful conditions like Infection , Carnitine Deficiency etc.
- Partial deficiency has also been noted in Organic Acidemias involving Carboxylases enzymes or other Organic Acidemias e.g. PA / IVA / MMA.
- Biotinidase deficiency is an autosomal recessively inherited disorder, which is now days considered as treatable by Tab. Biotin supplementation.
- In cases of partial deficiency it is always necessary to rule out other IEMs by MS/MS and GC/MS.
- Neonatal screening for this disorder has also been recommended routinely.

Dr. Anil Jalan
MD DCH MCPS

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Carnitine / Acyl Carnitine Profile – Derivatization method - DBS

Date: 01/10/2022

Sample: Dried Blood spots

Method: Derivatization method

Instrument: LC/MS/MS (Agilent LC6420)

	Carnitine / Acyl Carnitine		Normal Range (uMol/L)
1	Total Carnitine	34.37	20 – 87.7
2	Free Carnitine	18.24	24.7 – 66.6 (<8.6 for CTD)
3	Acyl Carnitine	16.13	
4	Free / Acyl ratio	1.13	> 2.0
5	Acyl Carnitine Profile	Normal acyl carnitine profile	
	Amino acids		Normal Range (uMol/L)
1	Alanine	375.35	650.0
2	Glycine	343.88	1000.0
3	Leucine - Isoleucine	190.34	300.0
4	Valine	122.05	300.0
5	Methionine	34.16	75.0
6	Phenylalanine	111.52	150.0
7	Tyrosine	42.52	400.0
8	Arginine	23.06	70.0
9	Citrulline	16.90	55.0
10	Ornithine	268.52	393.0
11	Glutamic acid	336.90	1070.0
12	Aspartic acid	30.40	737.0

Impression: Normal total carnitine, low free carnitine and low free / acyl carnitine ratio. Normal levels of amino acids.

Adv: Please correlate clinically.

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Carnitine / Acyl Carnitine Profile – Derivatization method - DBS

	Acyl Carnitines	Values	Cut-offs (uMol/L)	Ratios	Values	Cut-Offs
1	C 2	9.40	25.0	C3/C0		0.20
2	C 3	1.00	6.00	C3/C2		0.20
3	C 3DC	0.07	0.30	C3/C16		3.50
4	C 4	0.31	1.30	C4/C2		0.10
5	C 4-OH	0.14	0.65	C4/C3		0.70
6	C 4DC	0.05	0.63	C4OH/C2		0.026
7	C 5	0.19	0.70	C5/C2		0.05
8	C 5-OH	0.09	0.90	C5-OH/C2		0.04
9	C 5:1	0.02	0.25	C5:1/C2		0.20
10	C 5 DC	0.04	0.35	C5DC/C2		0.02
11	C 6	0.05	0.45	C5DC / C12		
12	C6DC	0.05	0.87			
13	C 8	0.07	0.45	C8/C2		0.03
14	C8:1	0.03	1.50			
15	C 10	0.08	0.45	C10:1/C2		
16	C 10: 1	0.06	0.30			
17	C 10: 2	0.02	0.15			
18	C 12	0.11	0.50			
19	C 14	0.16	0.80	C14/C2		0.03
20	C14OH	0.01	0.04			
21	C 14: 1	0.17	0.60	C14:1/C16		0.30
22	C14 : 2	0.03	0.15			
23	C 16	2.15	7.50			
24	C16:1	0.16	0.35			
25	C16: 1OH	0.01	0.14			
26	C 16-OH	0.02	0.15	C16-OH/C2		0.01
27	C 18	0.47	2.50	C18 :1/C2		0.15
28	C 18-OH	0.01	0.10			
29	C 18: 1	1.03	3.50	C18-OH/C2		0.10
30	C 18: 1-OH	0.01	0.16	C18:1OH/C2		0.01
31	C18:2	0.10	0.90			
32	C16DC	0.01	0.10			
33	C18:1DC	0.01	0.06			

Imp: Normal acyl carnitine profile.

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MD DCH MCPS

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Note:

1. Metabolic lab reports should not be interpreted in isolation and should always be correlated with clinical findings and with other medical reports
2. The results are to be used as guidance by medical professionals only.
3. The results provided above are semi-quantitative as they are performed on dried blood spots and they relate only to the sample provided. In case of clinical suspicion of any specific disease, quantitative assays may be performed on liquid blood samples.
4. Additional studies including plasma amino acids, urine organic acids and quantitation of metabolites may be needed in case child develops any symptoms related to metabolic disorders.
5. Certain disorders may develop slowly or may have intermittent elevation of certain metabolites.
6. Some values may be altered by drugs received by the patient prior to sampling. This should be considered while interpreting the lab reports
7. Normal amino acids and acyl carnitine profile does not rule out all Inborn Errors of Metabolism
8. The above method is adapted from method described by Rinaldo P et al. (Rinaldo P, et al. Acylcarnitine profile analysis. ACMG standards and Guidelines. Genetics in Medicine, 2008. Vol 10 No 2.)
9. The above results may be published only after approval from the authorized signatory.

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AMINO ACIDS – Plasma – UHPLC

Date: 30/09/2022

Sr. No	Test	Plasma (μmol/L)	Reference range (μmol/L)
1	Aspartic Acid	68	34 – 94
2	Glutamic Acid	33	17 – 69
3	Asparagine	37	28 – 65
4	Serine	138	92 – 196
5	Glutamine	300	457 – 857
6	Histidine	116	68 – 108
7	Glycine	442	166 – 450
8	Threonine	57	102 – 246
9	Citrulline	7	19 – 52
10	Alanine	294	242 – 594
11	Arginine	38	1 – 81
12	Tyrosine	46	35 – 107
13	Cystine	4	36 – 58
14	Valine	142	155 – 343
15	Methionine	27	13 – 60
16	Tryptophan	13	10 – 140
17	Phenylalanine	60	34 – 86
18	Iso – Leucine	24	34 – 106
19	Ornithine	17	47 – 195
20	Leucine	146	86 – 206
21	Lysine	115	116 – 276
22	Proline	106	58 – 324
23	Total Amino acids	2230	3000 - 5000
24	Total BCAA	312	< 600
25	Gly / BCAA	1.41	
26	Ala / BCAA	0.94	
27	Gly / Total	20%	< 10 %
28	Ala/ Total	13%	< 10%

Imp: No significant abnormality detected.

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FIBROBLAST GROWTH FACTOR 21

Date: 03/10/2022
 Sample: Serum
 Method: ELISA

Sample	Observed Values	Normal Controls Reference Range
B/o Praveena	780.49 pg/ml	Children: 30 – 255 Adults < 50 yrs : 140 – 411 Adults > 50 yrs: 177 – 499
Normal Control	153.90 pg/ml	

Impression: Elevated level of serum FGF21.

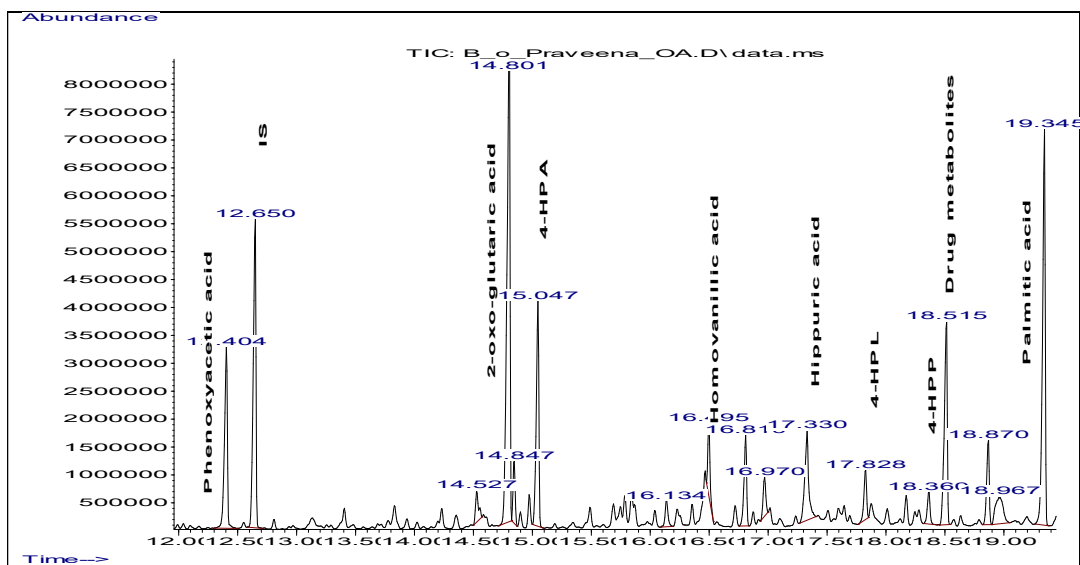
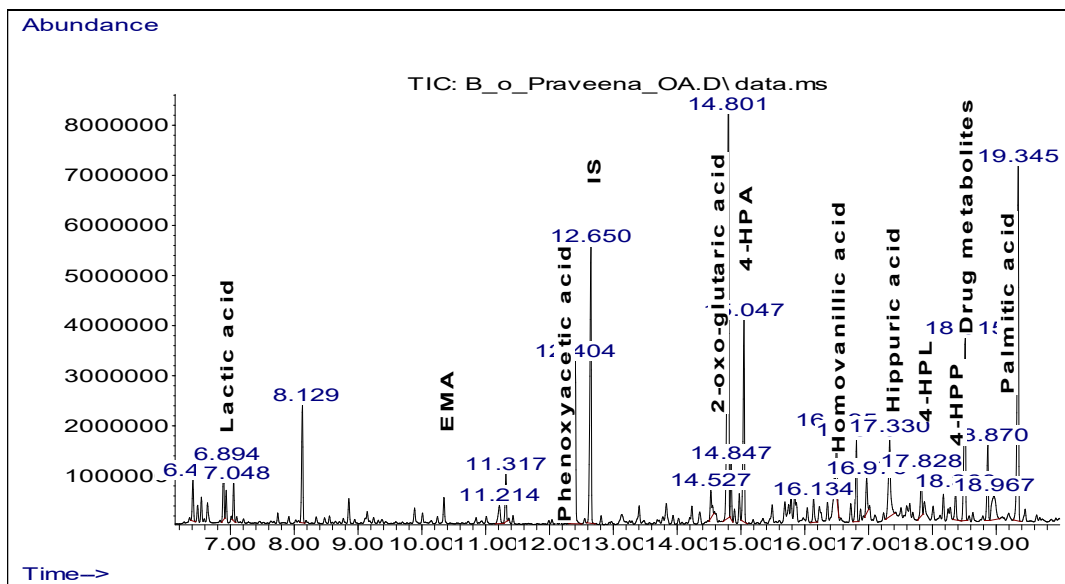
Note:

1. FGF21 is elevated in Mitochondrial Diseases, specially affecting muscles.
2. It increases with age and is related to aging process
3. It is also increased in disordered metabolic states like Obesity, type 2 Diabetes Mellitus and Cardiovascular Diseases

Dr. Anil B Jalan
 MD DCH MCPS
 Pediatric Geneticist

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GC-MS report of Organic Acid Analysis- Reviewed Report



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GC-MS report of Organic Acid Analysis- Reviewed Report

RT	Compound
6.945	Lactic acid
10.355	Ethylmalonic acid (EMA)
12.404	Phenoxyacetic acid
12.650	ISTD
14.801	2-oxo-glutaric acid
15.047	4-hydroxyphenylacetic acid (4-HPA)
16.495	Homovanillic acid
17.330	Hippuric acid
17.828	4-hydroxyphenyllactic acid (4-HPL)
18.360	4-hydroxyphenylpyruvic acid (4-HPP)
18.515	Drug metabolites
19.345	Palmitic acid

Interpretation: Normal urine organic acid profile. No sign of any specific organic or dicarboxylic aciduria.

Adv.: Please correlate with:

- 1) Carnitine / acyl carnitine profile in blood
- 2) Plasma amino acids

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Date: 04/10/2022

To
Dr. D V Suresh
Salem

Dear Sir / Madam

Many thanks for referring samples of B/o Praveena with moderate PAH, reduced tone in UL and LL. As per your request we have completed the metabolic studies and following are the results:

1. Serum Biotinidase enzyme: Normal
2. TMS of blood: shows low free carnitine and low free / acyl carnitine ratio
3. GCMS of urine: shows no specific abnormality
4. Plasma amino acids: shows no specific abnormality
5. Serum FGF21: Elevated

The above results do not indicate any specific Inborn Error of Metabolism. I would like to suggest the following:

1. Serum ammonia, lactate, sugar to be done locally on fresh blood sample
2. Supplement Carnitine and CoQ
3. Genetic studies are strongly recommended
4. If you wish I can see this child during my next visit to Coimbatore- 7th- 9th November, 2022

Thanking you

Dr. Anil Jalan
MD DCH MCPS